

Programming Fundamental

Lab 11

objective:

- To understand the fundamental difference between stack and heap memory in program execution.
- To demonstrate how variables are stored in stack vs. heap memory.
- To implement an array using heap memory and observe how it regrows as more elements are inserted.

Task 1: Find Maximum from File

Read an integer array from a file. The first number indicates the size. Allocate the array dynamically on the heap. Find and display the maximum number.

Sample File (data.txt):

10 5 12 3 55 9 22 1 6 17 4

Sample Output:

Maximum Number: 55

Task 2: Search Number in File Array

Read an integer array from a file using heap allocation. Ask the user to enter a number. Search for the number using a separate function (passed via stack). Display if found or not.

Sample File (data.txt):

8 11 24 36 42 17 29 5 66

Sample Input:

Enter number to search: 42

Sample Output:

Number found!

Task 3: Square of Numbers

Read numbers from file into a heap array. Create a regrowable output array that stores the square of each number. Output the squared values.

Sample File (data.txt):

6 2 4 6 8 10 12

Sample Output:

Original Data: 2 4 6 8 10 12

Squared Data: 4 16 36 64 100 144

Task 4: Replace Values by Subtracting from Max

Use heap allocation for reading array. Use a function to find max. Subtract each value from max and replace it in array.

Sample File (data.txt):

5 3 6 2 9 4

Sample Output:

Original Data: 3 6 2 9 4

Modified Data: 6 3 7 0 5

Task 5: Remove Number

Read data from file into a heap array. Ask user for a number to remove. Swap it with the last element and replace last element with 0.

Sample File (data.txt):

7 3 5 7 9 11 13 15

Sample Input:

Enter number to remove: 9

Sample Output:

Modified Data: 3 5 7 15 11 13 0

Task 6: Separate One-digit and Two-digit

Use regrowable arrays on heap to store one-digit and two-digit numbers separately from a file.

Sample File (data.txt):

10 1 23 9 11 6 45 2 89 7 10

Sample Output:

One-digit Numbers: 1 9 6 2 7

Two-digit Numbers: 23 11 45 89 10

Task 7: Word Lengths from File

Read words from a file and store the length of each word in a regrowable array.

Sample File (text.txt):

HELLO WORLD CODE 12345

Sample Output:

Word Lengths: 5 5 4 5

Task 8: Separate Even and Odd Numbers

Read integers from file into a heap array. Separate into even and odd using two regrowable arrays.

Sample File (data.txt):

12 4 7 6 3 10 11 14 13 8 5 2 1

Sample Output:

Even Numbers: 4 6 10 14 8 2

Odd Numbers: 7 3 11 13 5 1

Task 9: Sort Even Ascending & Odd Descending order

From heap-allocated array, separate even and odd numbers. Sort even in ascending order and odd in descending order. Recombine the array.

Sample File (data.txt):

10 9 4 6 1 3 8 5 2 7 10

Sample Output:

Updated Data: 2 4 6 8 10 9 7 5 3 1

Task 10: Shift Even Left, Odd Right

Read integers into heap array. Move even numbers to the left (ascending) and odd numbers to the right (descending).

Sample File (data.txt):

10 1 4 7 2 9 8 3 6 5 10

Sample Output:

Modified Data: 2 4 6 8 10 9 7 5 3 1

Task 11: Shift and Remove Vowel

Read character text into a heap-allocated array. Shift left n times (where n = text length). On each shift, if last char is vowel, remove it and shrink the array manually.

Sample File (text.txt):

abcdefghijklm

Sample Output (after vowel removal):

Modified Text: bcdgjkbcdbcd

Task 12: Rearrange Words

Read space-separated words from file into a heap array. Separate words into digit-only and alphabet-only using regrow arrays. Rearrange so digit words come first.

Sample File (text.txt):

123 45 HELLO 9999 CODE 8888 TEST

Sample Output:

Modified Text: 123 45 9999 8888 HELLO CODE TEST